

Applied Chemistry
Assignment_1-MID TERM 1

1. Write the structural unit and two important applications of the following: (3)
 - i) Nylon 6,6.
 - ii) Urea-formaldehyde
 - iii) PVC
2. What is ultra-thin molecular weight polythene? Write their unique properties and applications. (3)
3. Define conducting polymer, their types and applications of conducting polymers in engineering. (6)
4. Describe the method of preparation of polymers, Nylon 6,6. (2)
5. Describe the method of preparation and properties and application of the following: (Any two) 6
 - i) Teflon (PTFE)
 - ii) Polyacrylonitrile (PAN)
 - ii) Bakelite
 - iv) Glyptal
6. What are the condensation and polymerisation processes? Explain with example. 6
7. Explain the difference between addition and condensation polymerization. 2
8. Discuss the mechanism of free radical and cationic polymerization using suitable examples. 2
- 9.

(a) A polymer has been found to possess the population of various molecules as follows: (5)

- (i) 5 molecules of molecular mass each 25000.
- (ii) 15 molecules of molecular mass each 20000.
- (iii) 10 molecules of molecular mass each 45000.
- (iv) 25 molecules of molecular mass each 62000.
- (v) 10 molecules of molecular mass each 100000.

Calculate its Number Average Molecular weight, Weight Average Molecular weight and P.D.I. Explain the significance of PDI.

(b) How will you prepare Nylon-6 from cyclohexane? (2)

(c) Write the names and structures of the monomers for (i) Neoprene (ii) Teflon. Also mention their important applications of the above two polymers. (3)

(d) Write short notes on (any two):- (5)

- (i) Adhesives
- (ii) Polymer composites
- (iii) Thermoset and Thermoplastic polymers

10. Write down the reactions involved in synthesis, properties and uses of 5

i) Polycarbonates

ii) Nylon 6,6

11. Enumerate doped conducting polymers. What are methods of doping in conducting polymers.

12. List down the preparation, properties, and application Bakelite with chemical reaction involved. (3)

13. Write down a note on condensation polymerization and co-polymerisation giving examples of both. (2)
14. How functionality of polymer affects the polymeric structure? (2)
15. Enumerate the difference between thermoplastic and thermosetting polymers. (2)
16. Write a short notes on biodegradable polymers and electroluminescent polymers. (4)
17. Describe briefly the difference between LDPE and HDPE. Is LLDEP is a co-polymer? (2.5)
18. What are biodegradable synthetic polymers? Give some typical medical applications of these polymers. (3)
19. What is the functionality of (i) Phenol and (ii) ethylene. Which among them will yield a linear polymer? (3)
20. Write properties and uses of one of (i) Nylon 11 and (ii) Bakelite. (2)